

1. Which of the following is the primary site of gaseous exchange in the human lungs?

- A) Bronchi
- B) Bronchioles
- C) Alveoli
- D) Trachea

2. Which respiratory defense mechanism is most directly impaired by the cytotoxic chemicals in cigarette smoke?

- A) Mucociliary clearance (ciliary action).
- B) Cough reflex.
- C) Alveolar macrophage activity.
- D) Sneezing reflex.
- E) Both A and C.

3. Which toxic component of cigarette smoke binds to hemoglobin with a much higher affinity than oxygen, reducing blood oxygen-carrying capacity?

- A) Tar.
- B) Nicotine.
- C) Carbon monoxide.
- D) Formaldehyde.

4. Which of the following chronic obstructive pulmonary diseases (COPD) is characterized primarily by the permanent destruction and enlargement of the alveoli?

- A) Chronic bronchitis.
- B) Emphysema.
- C) Asthma.
- D) Pneumonia.

5. Why do chronic smokers often develop a persistent "smoker's cough"?

- A) The smoke stimulates the rapid growth of new cilia.
- B) The cilia are paralyzed or destroyed, so the body relies on coughing to clear excess mucus.
- C) The smoke creates excess lung surfactant.
- D) The smoke triggers an increase in alveolar surface area.

6. Which cells are responsible for producing the sticky mucus that lines the conducting airways?

- A) Ciliated epithelial cells.
- B) Goblet cells.
- C) Macrophages.
- D) Chondrocytes.

7. Which structure prevents food and liquid from entering the trachea during swallowing?

- A) Pharynx
- B) Epiglottis
- C) Larynx
- D) Bronchus**

8. What is the primary function of the alveoli in the respiratory system?

- A) To produce vocal sounds
- B) To warm and moisten incoming air
- C) To facilitate the exchange of oxygen and carbon dioxide
- D) To trap dust and foreign particles

9. Which of the following is NOT a method the nasal cavity uses to condition inhaled air?

- A) Warming it through rich blood supplies
- B) Moistening it through mucus secretion
- C) Filtering out large particles using coarse hairs
- D) Absorbing excess oxygen into the blood stream

10. What occurs when the diaphragm contracts and moves downward?

- A) Exhalation, as chest volume decreases
- B) Inhalation, as chest volume increases
- C) The larynx closes to protect the vocal cords
- D) Gas exchange begins in the bronchioles

11. What are the tiny, hair-like structures that sweep mucus and trapped debris out of the respiratory tract called?

- A) Alveoli
- B) Surfactants
- C) Cilia
- D) Goblet cells

12. Which region serves as a passageway for both air and food?

- A) Trachea
- B) Pharynx
- C) Larynx
- D) Bronchi

13. What is the main function of the C-shaped cartilage rings found in the trachea?

- A) To produce the voice
- B) To actively exchange gases
- C) To prevent the airway from collapsing during inhalation
- D) To secrete pulmonary surfactant